

Take a clear photo of the motherboard inside your own or a friend's laptop/desktop (or find a labeled image online), and label the CPU socket, RAM slots, PCIe slots, and power connectors using any image editing tool.

Question:2

Pick any three popular CPUs used in gaming laptops or desktops (for example, Intel Core i5-12400F, AMD Ryzen 5 5600, Intel Core i7-13700K). Research and create a table comparing their clock speed (GHz), number of cores/threads, cache size (L2/L3), TDP (W), and current price in India.

Answer: 2

Comparison Table

CPU Model	Clock Speed (GHz)	Cores / Threads	Cache Size (L2/L3)	TDP (Watts)	Approximate Price in India (INR)
Intel Core i5-12400F	2.5 GHz (Base) / 4.4 GHz (Max)	6 Cores / 12 Threads	L2: 7.5 MB, L3: 18 MB	65W (Base) / 117W (Max)	₹9,500 - ₹10,500
AMD Ryzen 5 5600	3.5 GHz (Base) / 4.4 GHz (Max)	6 Cores / 12 Threads	L2: 3 MB, L3: 32 MB	65W	₹10,000 - ₹11,000
Intel Core i7-13700K	3.4 GHz (Base) / 5.4 GHz (Max)	16 Cores / 24 Threads	L2: 24 MB, L3: 30 MB	125W (Base) / 253W (Max)	₹32,000 - ₹34,000

Question:3

For each CPU in your comparison table, identify at least one compatible motherboard model (mention brand and model, e.g., ASUS PRIME B660M-A for Intel 12th Gen) and specify the socket type and form factor (ATX, mATX, ITX).

Answer: 3

Compatible Motherboards for the Selected CPUs

To make sure these processors work properly, they must be paired with a motherboard that has the exact same socket type. Below is the list of compatible motherboards for each CPU along with their form factors:

1. For Intel Core i5-12400F (Intel 12th Gen)

- **Motherboard Brand & Model:** MSI PRO B760M-A WIFI
- **Socket Type:** LGA1700
- **Form Factor:** Micro-ATX (mATX)
- **Note:** This board is budget-friendly, supports fast DDR5 or DDR4 RAM, and fits perfectly in standard PC cabinets.

2. For AMD Ryzen 5 5600 (AMD Zen 3)

- **Motherboard Brand & Model:** GIGABYTE B550M DS3H
- **Socket Type:** AM4
- **Form Factor:** Micro-ATX (mATX)
- **Note:** This is one of the most reliable and affordable motherboards for this Ryzen chip, providing great stability for budget gaming.

3. For Intel Core i7-13700K (Intel 13th Gen)

- **Motherboard Brand & Model:** ASUS ROG STRIX Z790-F GAMING WIFI
- **Socket Type:** LGA1700
- **Form Factor:** ATX (Full Size)

Question:4

Visit the official website of any one CPU you compared (Intel or AMD), and find the maximum supported RAM speed and PCIe version for that CPU. Write these details along with how they affect real-world performance in gaming or multitasking.

Hint: Look for 'Memory Types' and 'PCI Express Revision' in the CPU specs page.

Answer: 4

CPU Memory and PCIe Specs Analysis

I visited the official website of Intel to research the technical specifications of the [Intel Core i5-12400F](#) processor. Below are the findings regarding its maximum supported RAM speed and PCIe version, along with how they impact real-world system performance.

Official Technical Specifications:

- **Maximum Supported RAM Speed (Memory Types):** Up to **DDR5 4800 MT/s** or Up to **DDR4 3200 MT/s** (Depending on the motherboard used).
- **PCI Express (PCIe) Revision: PCIe 5.0 and 4.0** (Supports up to 20 total CPU PCIe lanes).

How These Specifications Affect Real-World Performance:

1. Impact of RAM Speed on Multitasking and Gaming:

- **Faster Multitasking:** Since this CPU supports DDR5 RAM speeds up to 4800 MT/s, the data moves much quicker between the memory and the processor. This allows me to open multiple heavy apps like Google Chrome, Discord, and a game at the same time without any system slowdown.
- **Higher FPS in Gaming:** High-speed RAM helps the CPU calculate game data (like physics and asset loading) much faster. In heavy esports games like *Valorant* or *Counter-Strike*, using faster DDR5 memory helps reduce stuttering and gives a significant boost to average frame rates (FPS).

2. Impact of PCIe Version on Storage and GPU:

- **Future-Proof Graphics (GPU):** With **PCIe 5.0** support, the motherboard can handle the latest and future high-end graphics cards at maximum bandwidth. It ensures that the GPU never encounters a data bottleneck while rendering heavy game visuals.
- **Ultra-Fast Storage (SSD):** The processor's **PCIe 4.0** lanes are used to connect modern NVMe M.2 SSDs. This drastically improves real-world performance by cutting down game loading screens to just a few seconds and making Windows boot up almost instantly.

